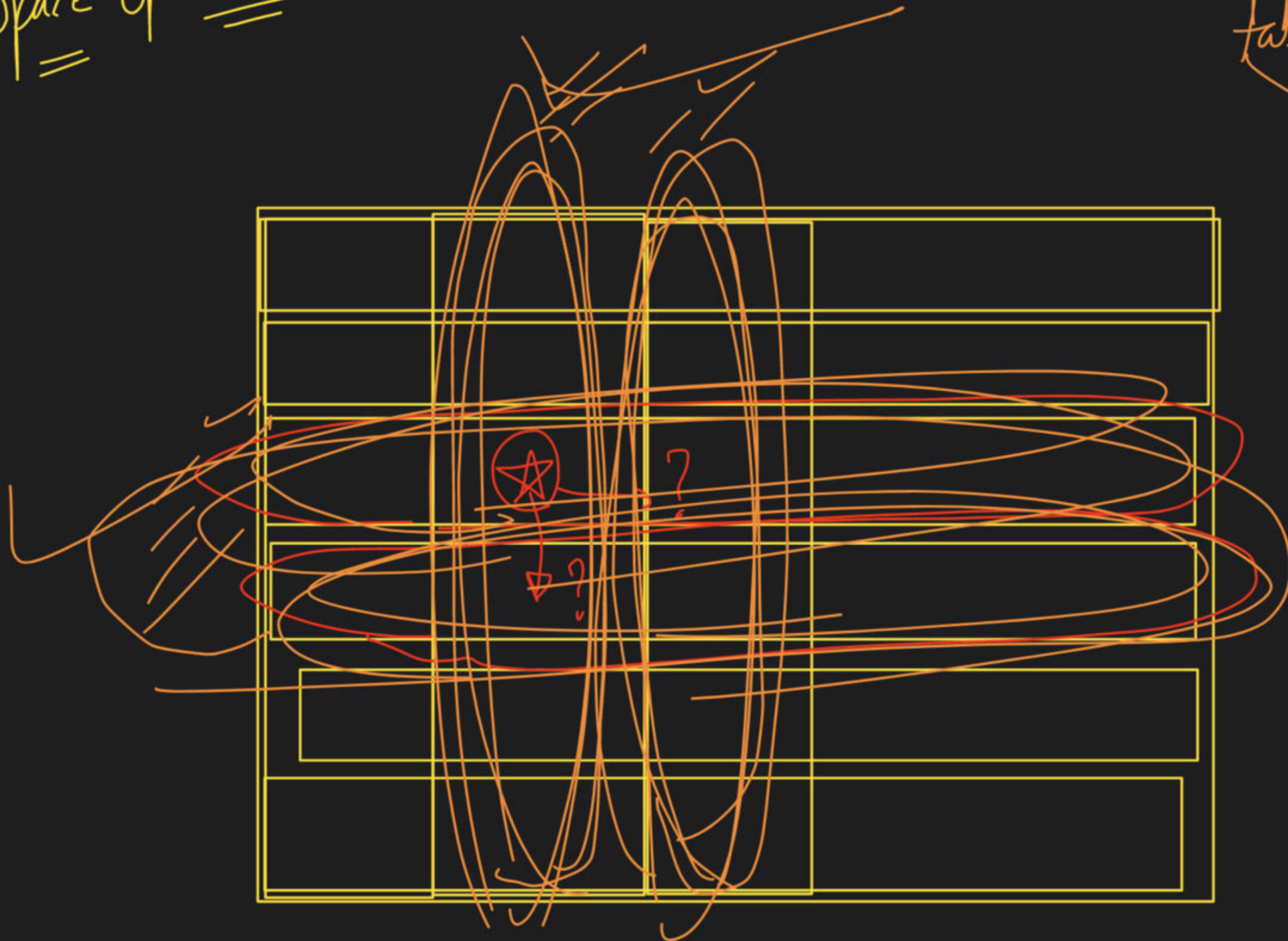


DP Class - 4 [Join here]

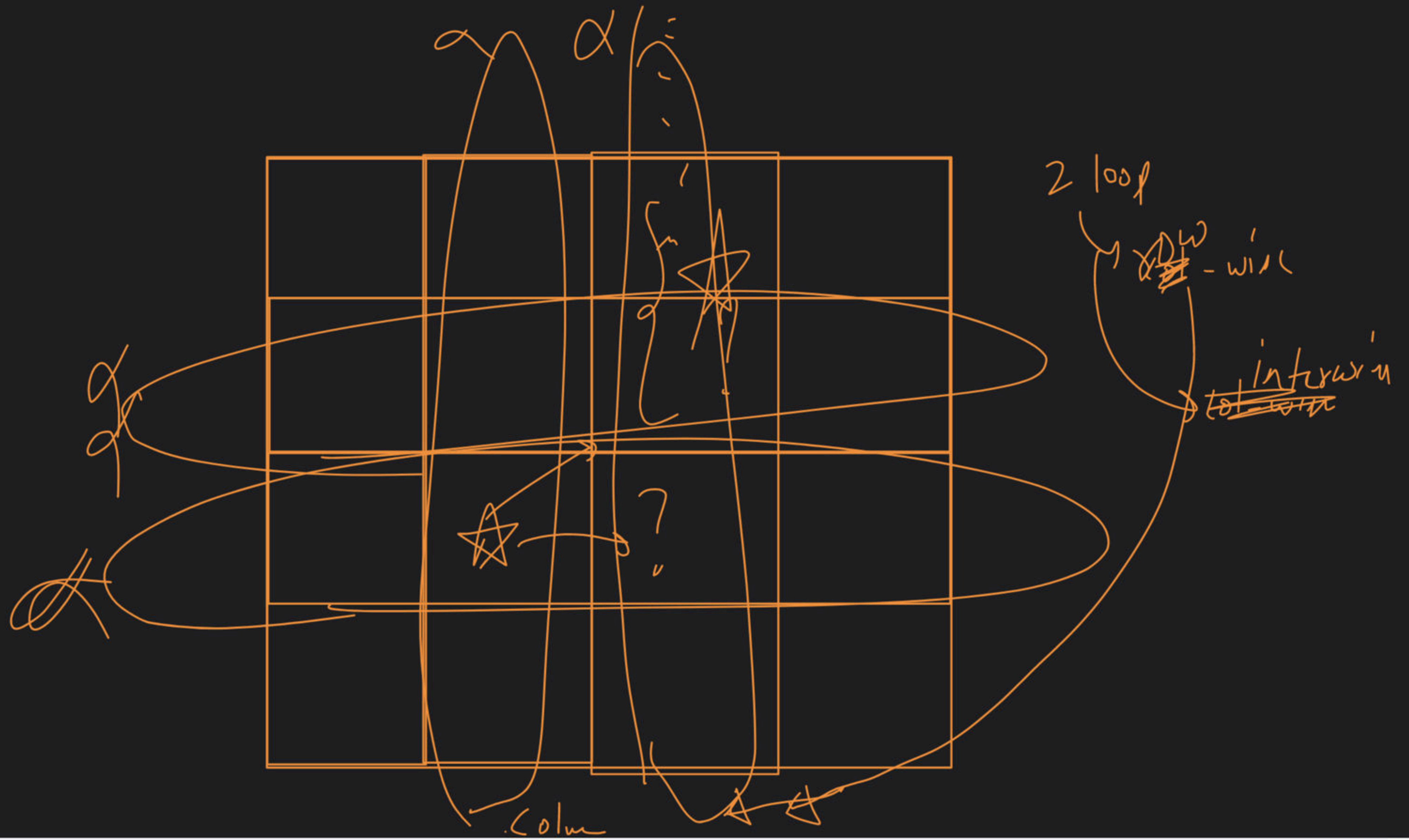
Special class

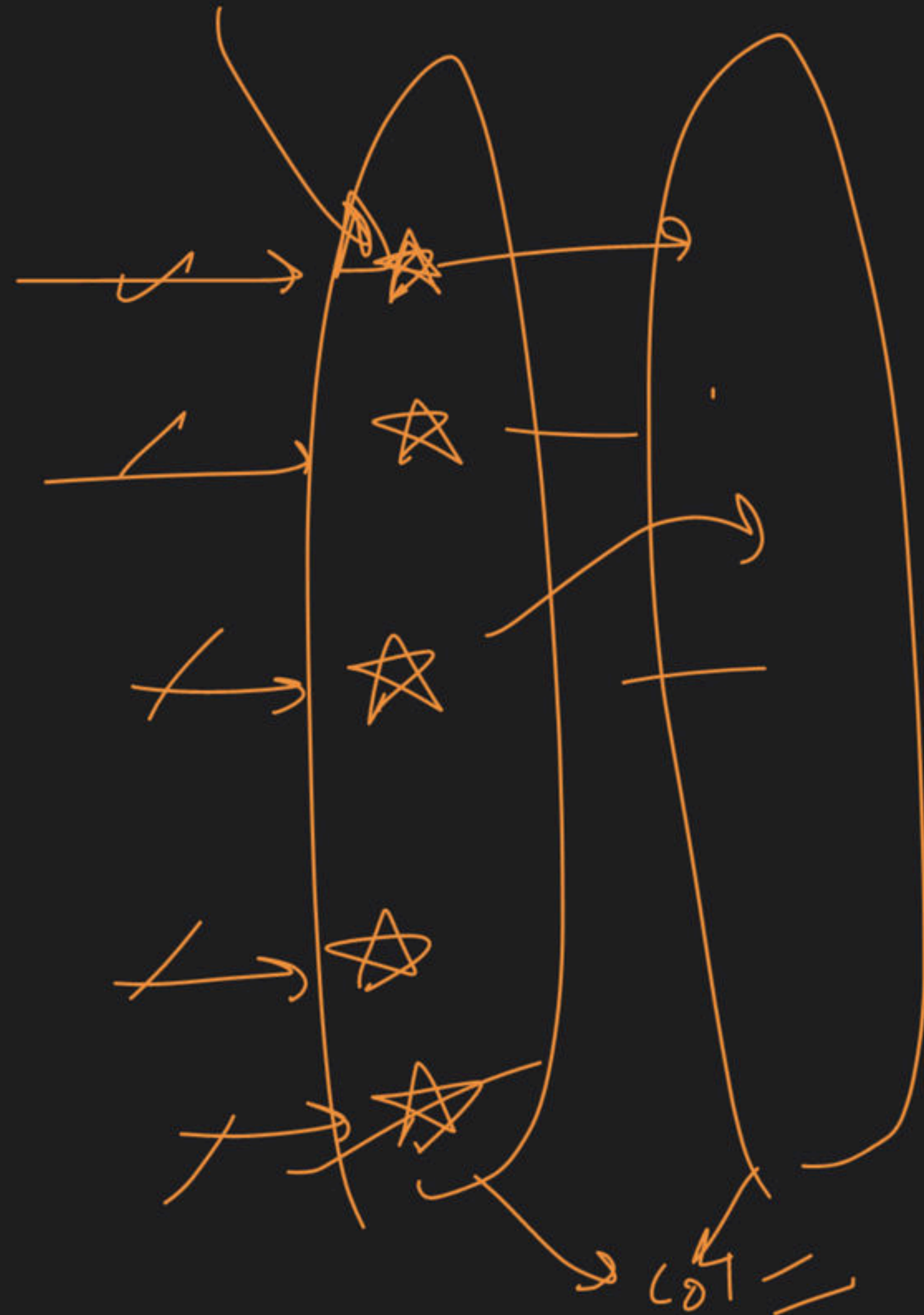
① Interchanged
=

Space Optimisation



tab → 2 loop
→ row-wise
→ col-wise



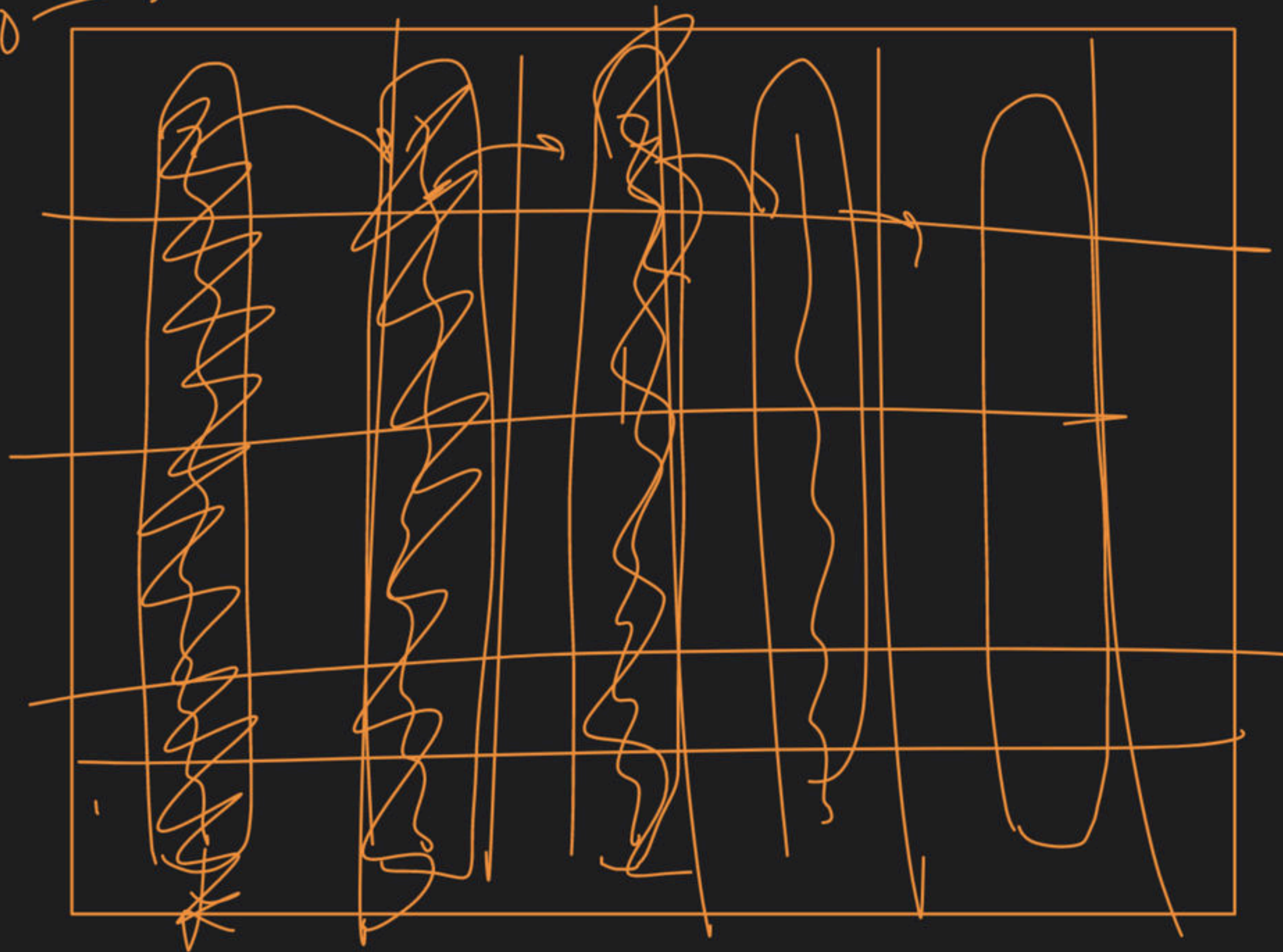


2 loop

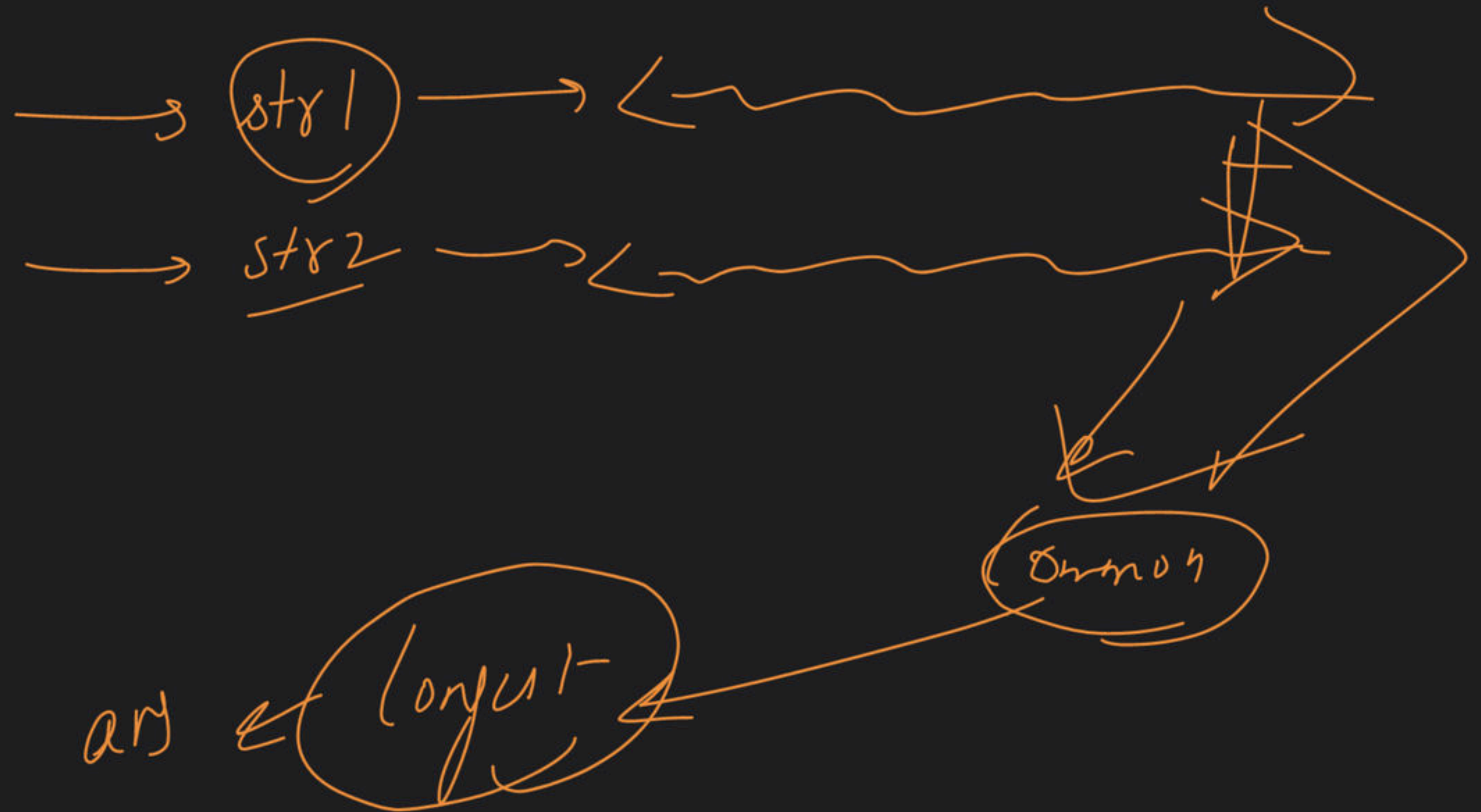
row win

interview

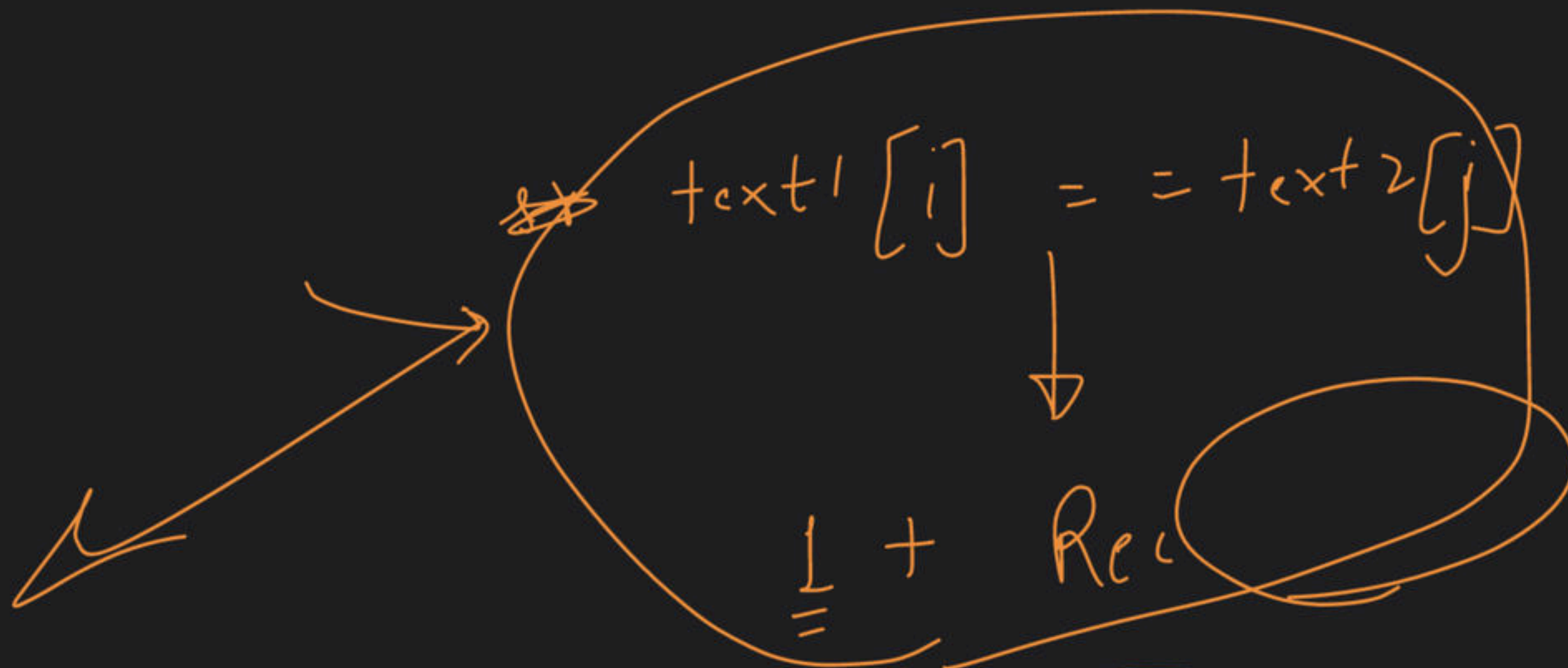
2d $\rightarrow$ $n \times n$ $\rightarrow$ tab (n^2)
S.O $\rightarrow n+n \rightarrow 2n \sim n //$



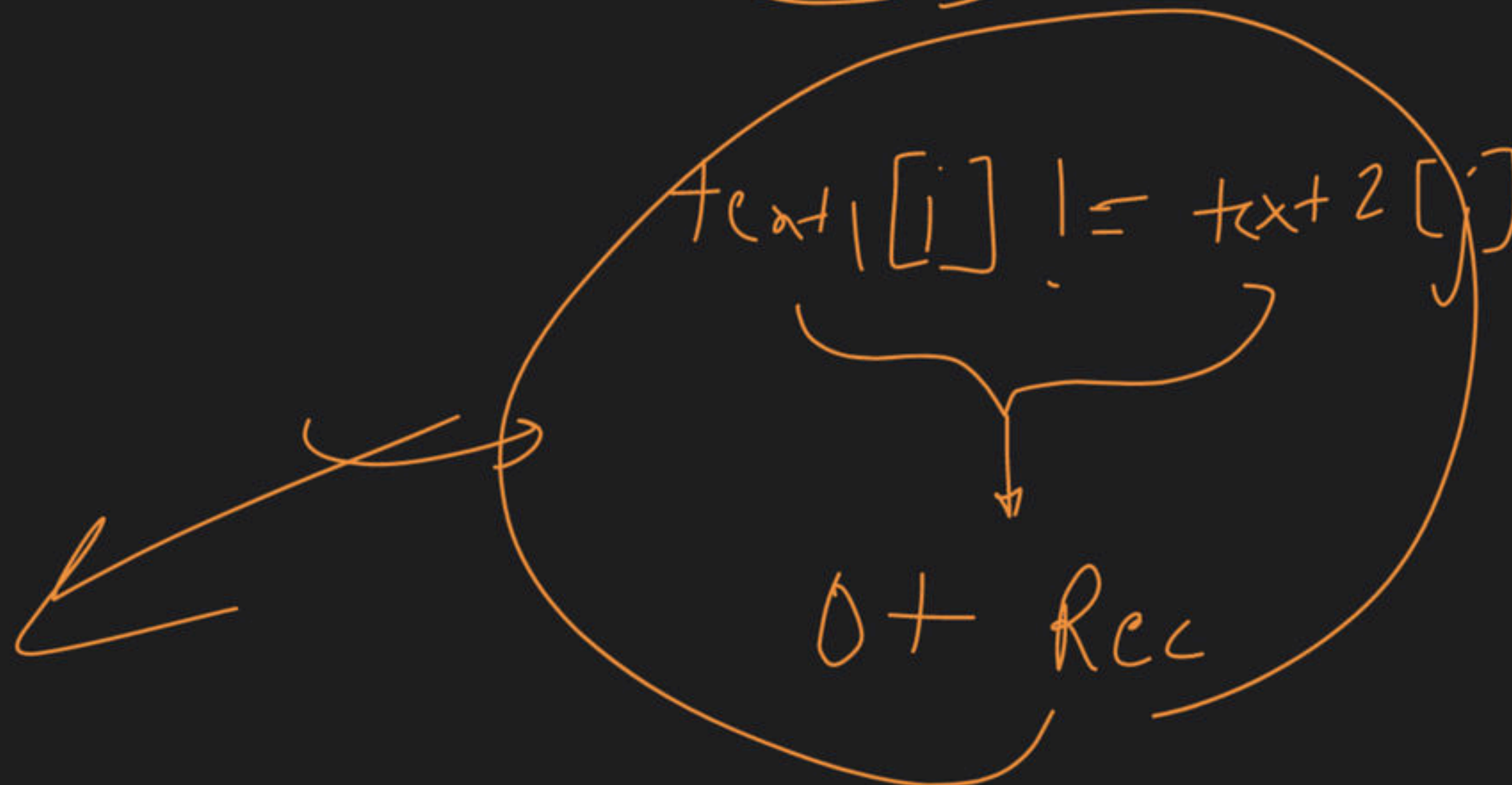
→ Longest Common Subsequence



abcd e
↑
i



a c e
↑
j



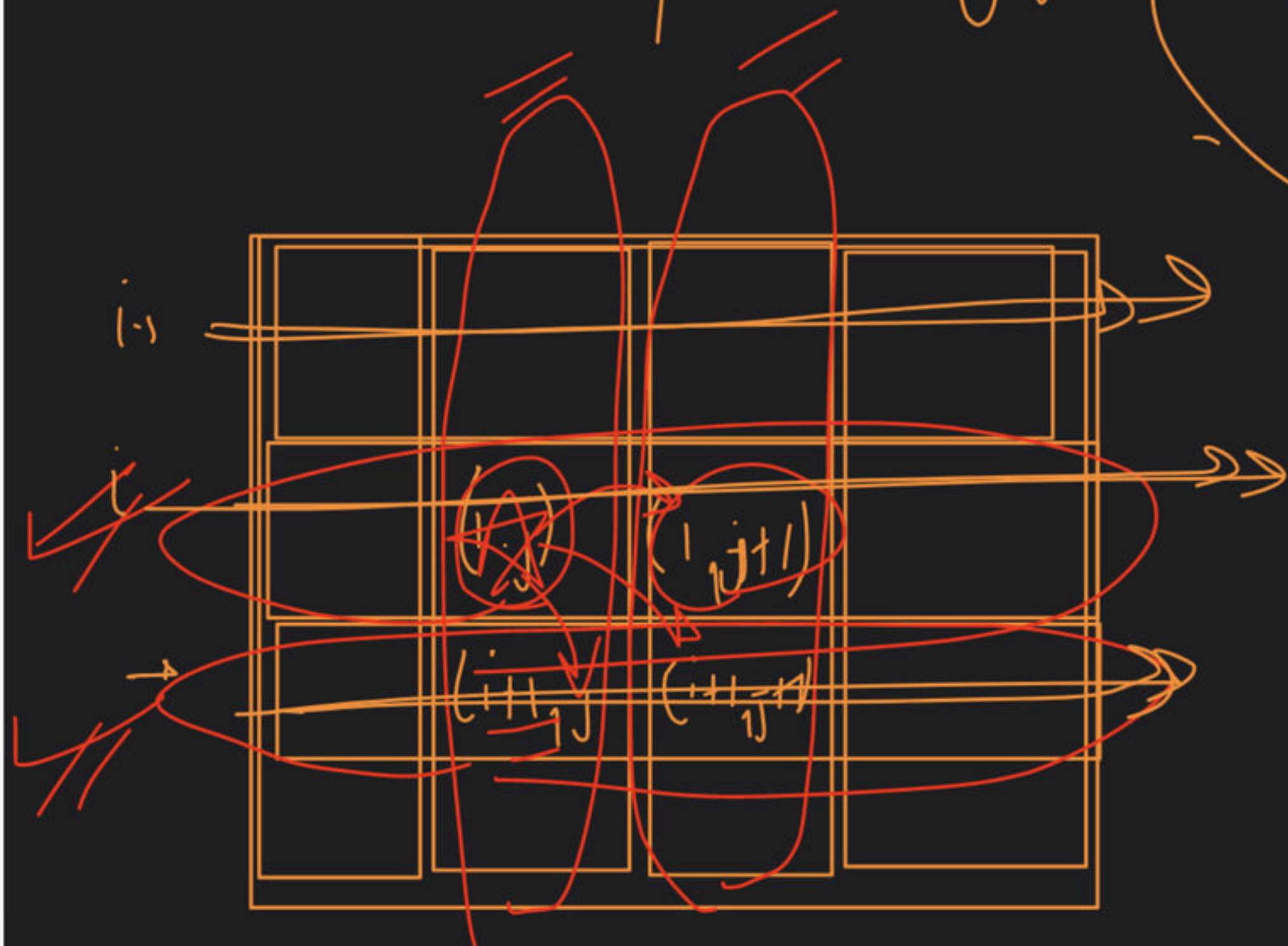
$dp[i][j]$

$dp[i+1][j+1]$

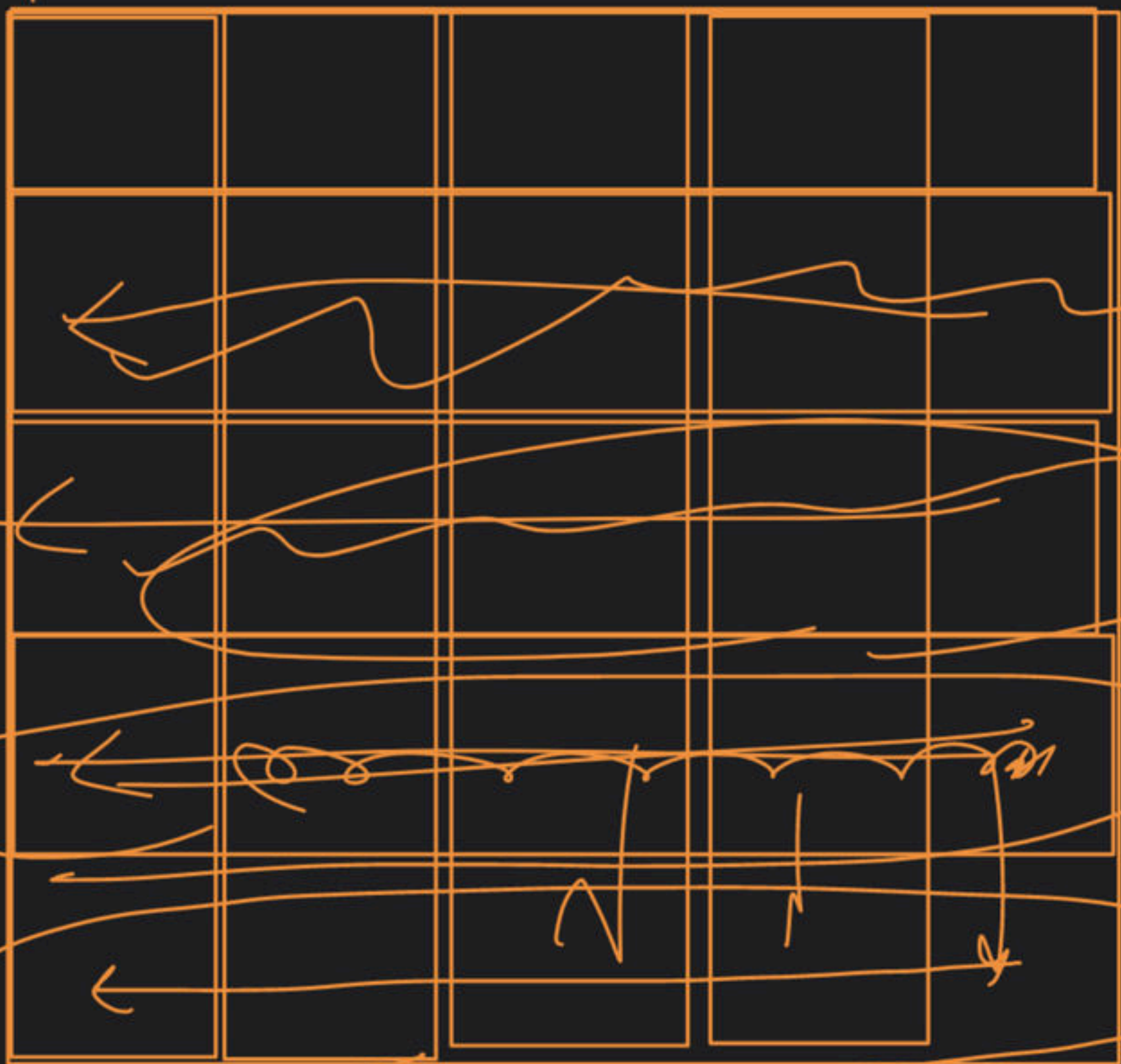
$dp[i+1][j]$

$dp[i][j+1]$

~~for loop ->~~



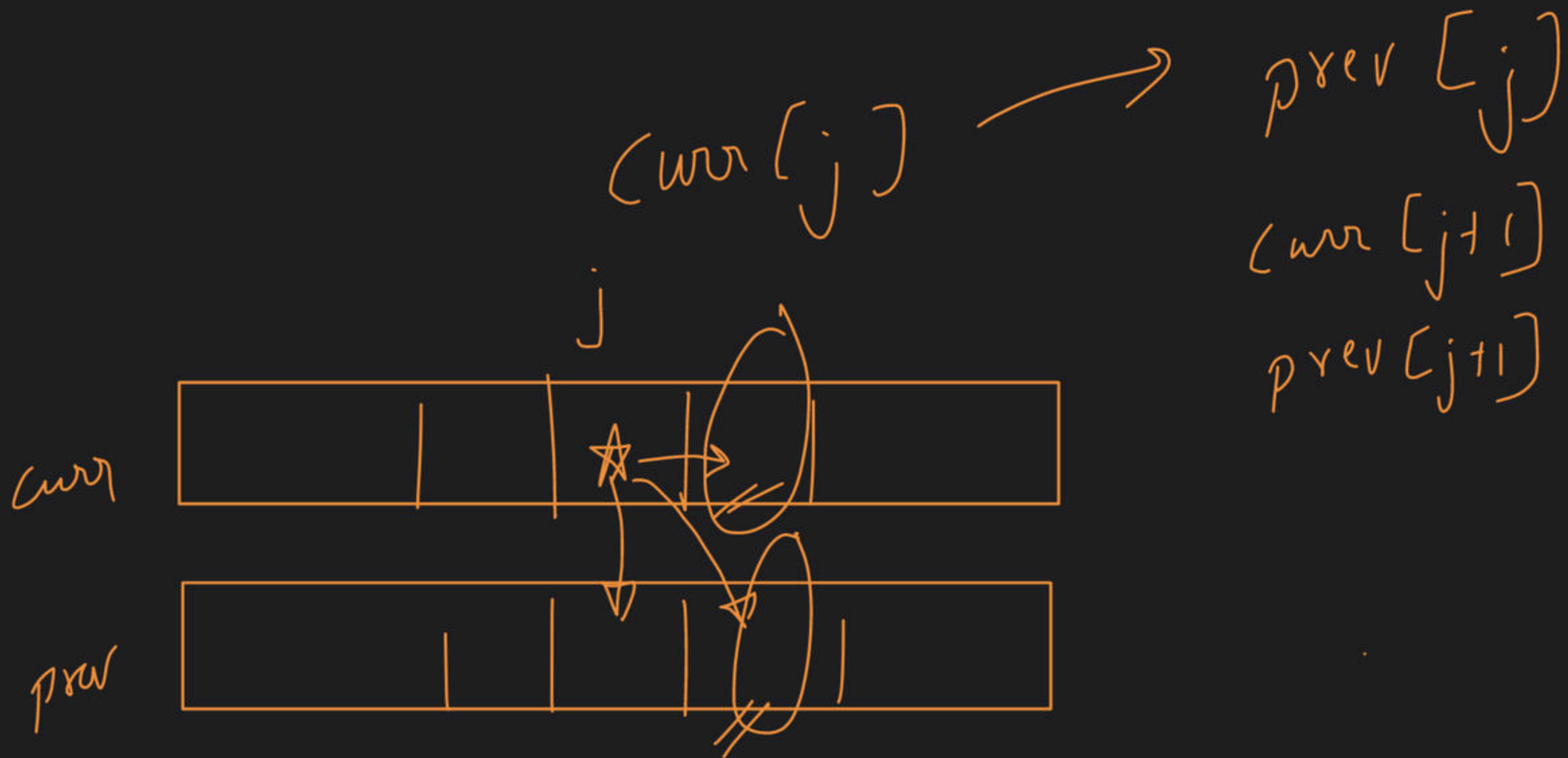
$k \times 2$ m

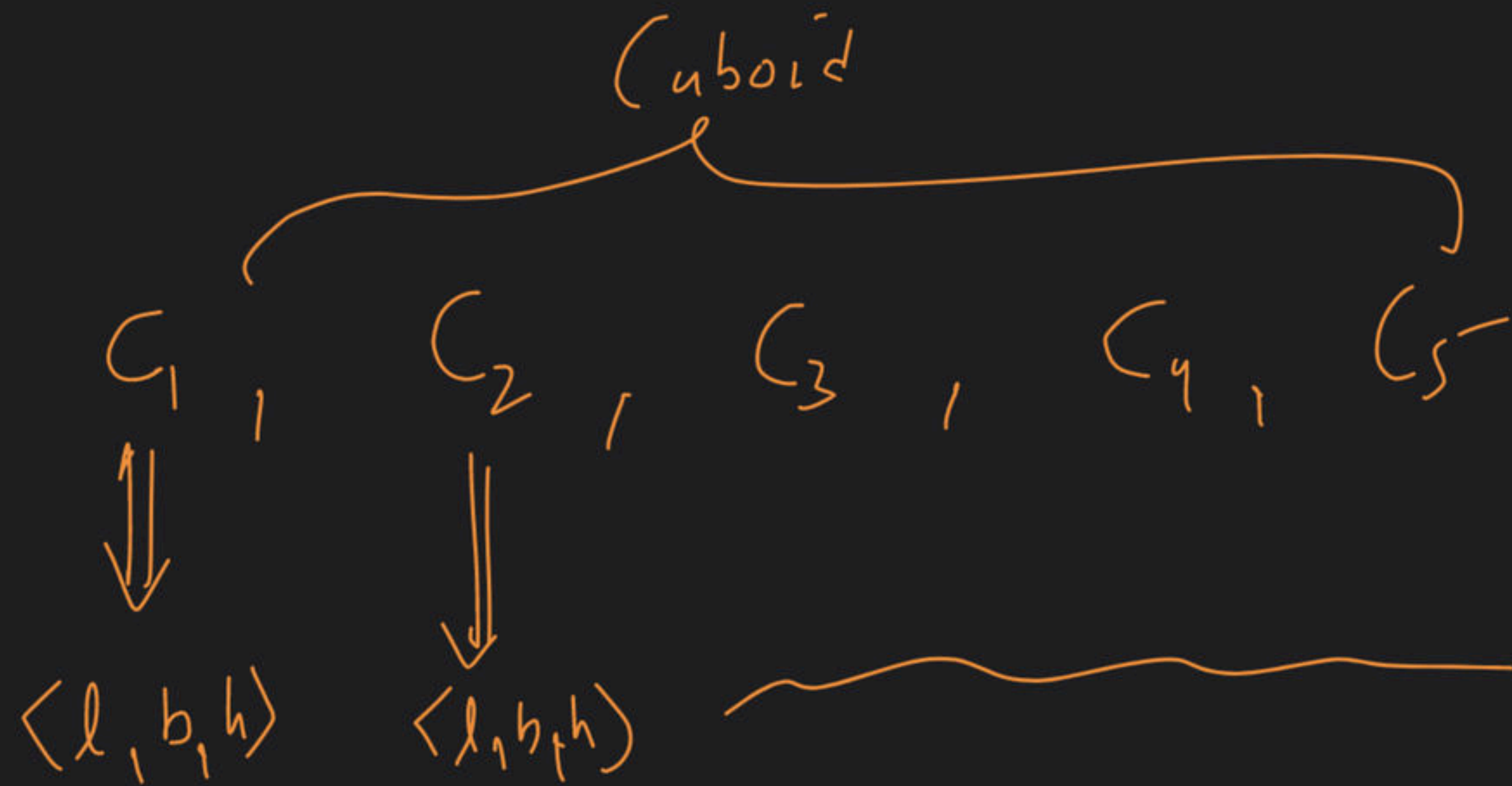


tall

$i-1$
 i
 $i+1$

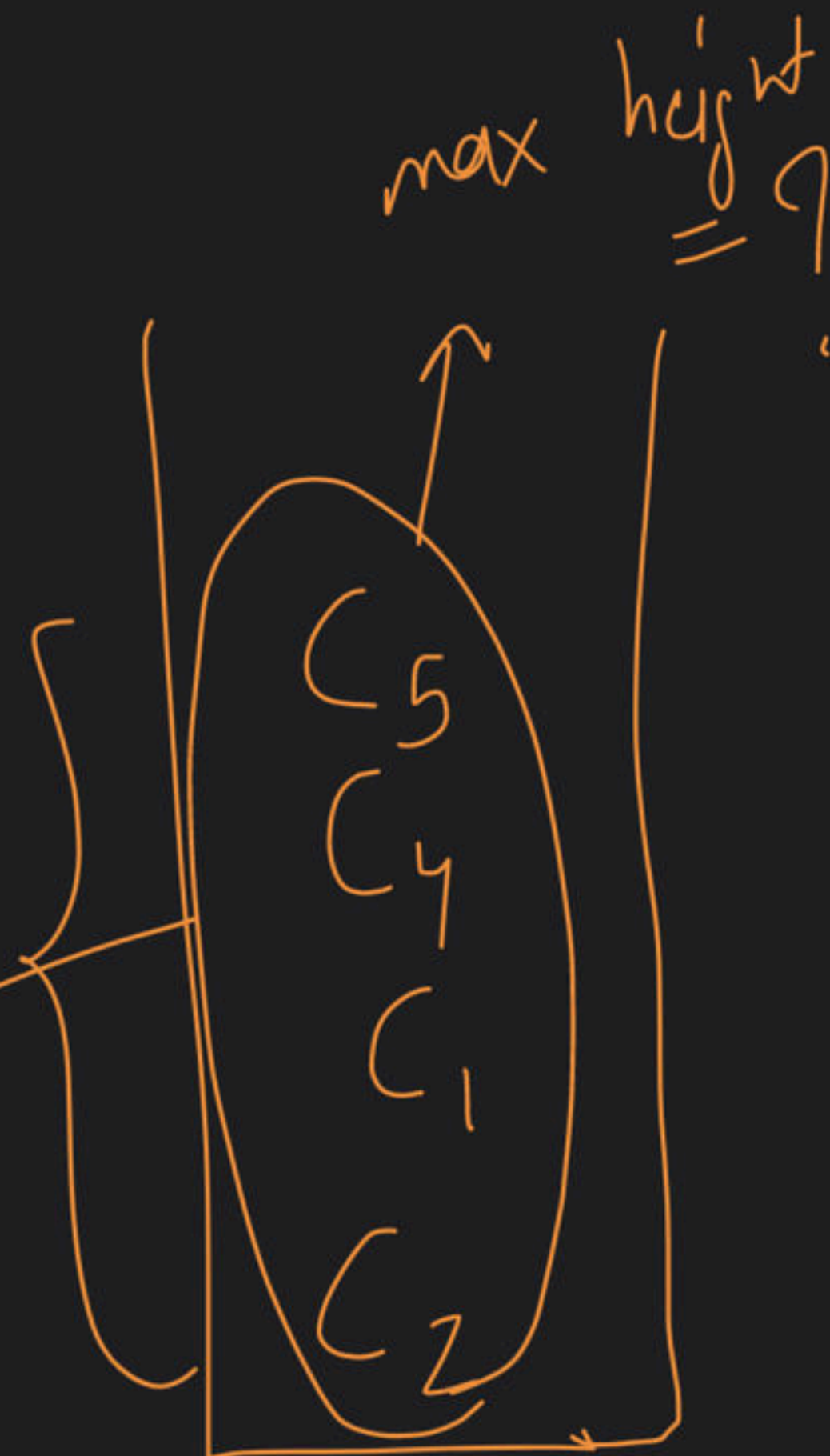
cur pos
~~not pos~~
~~pos~~



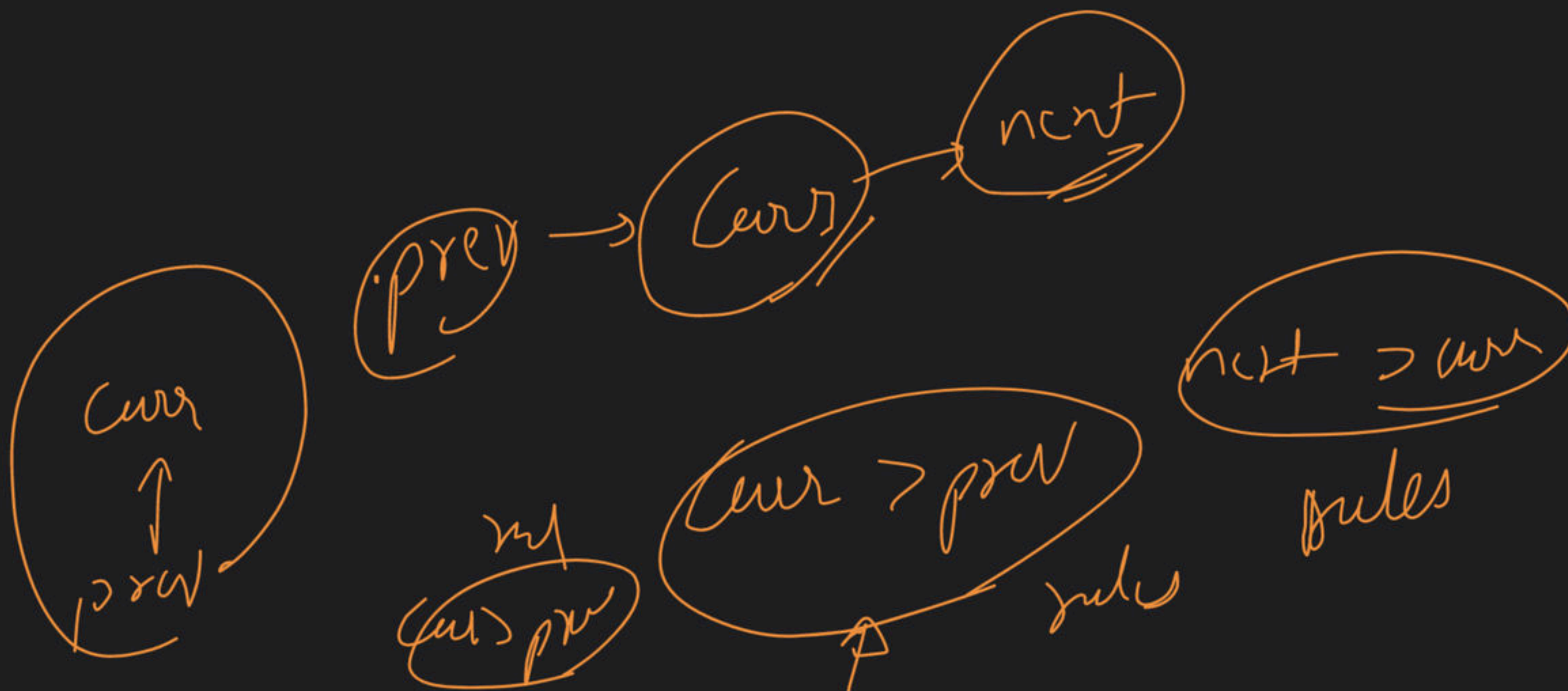


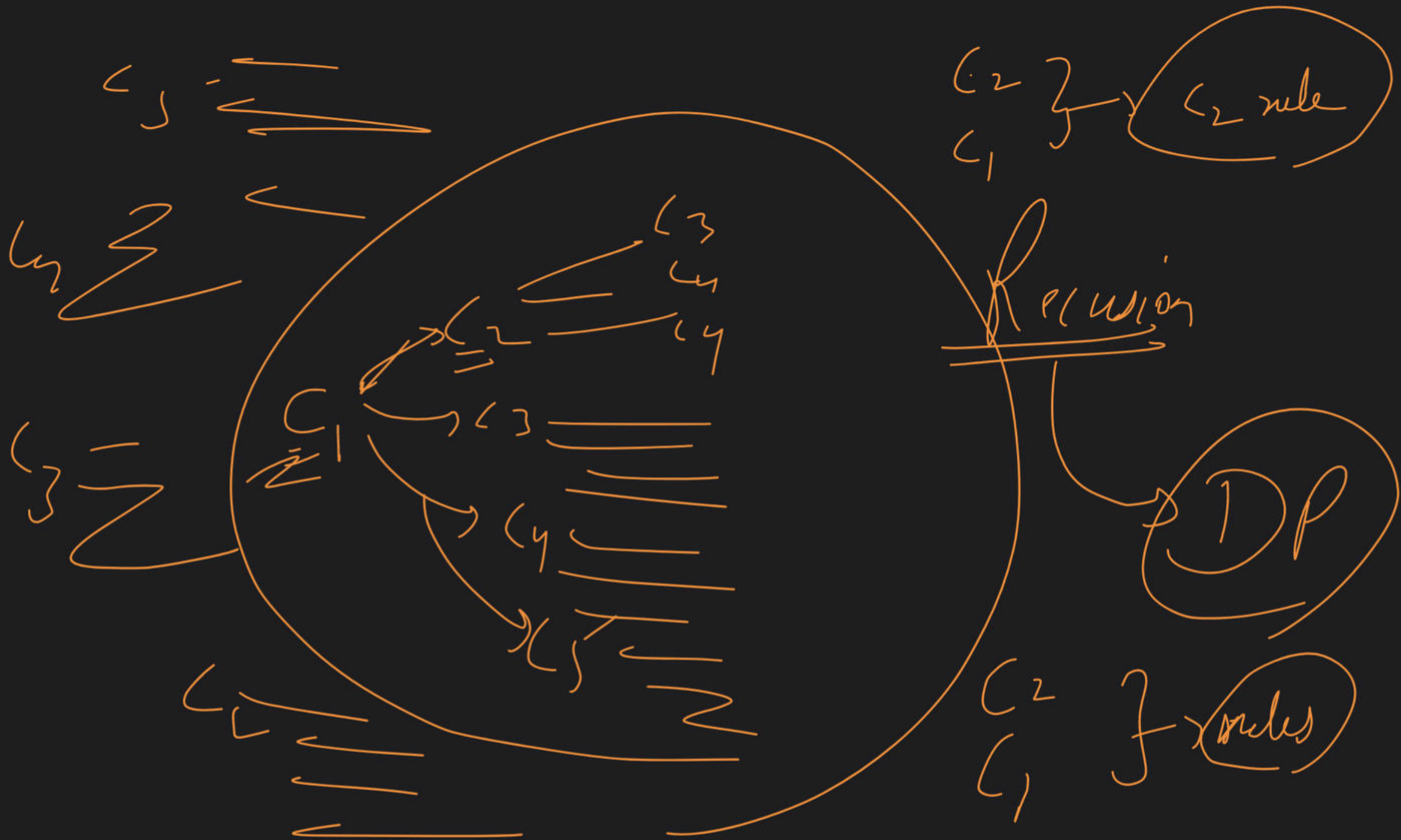
1 minutes

as per
rules



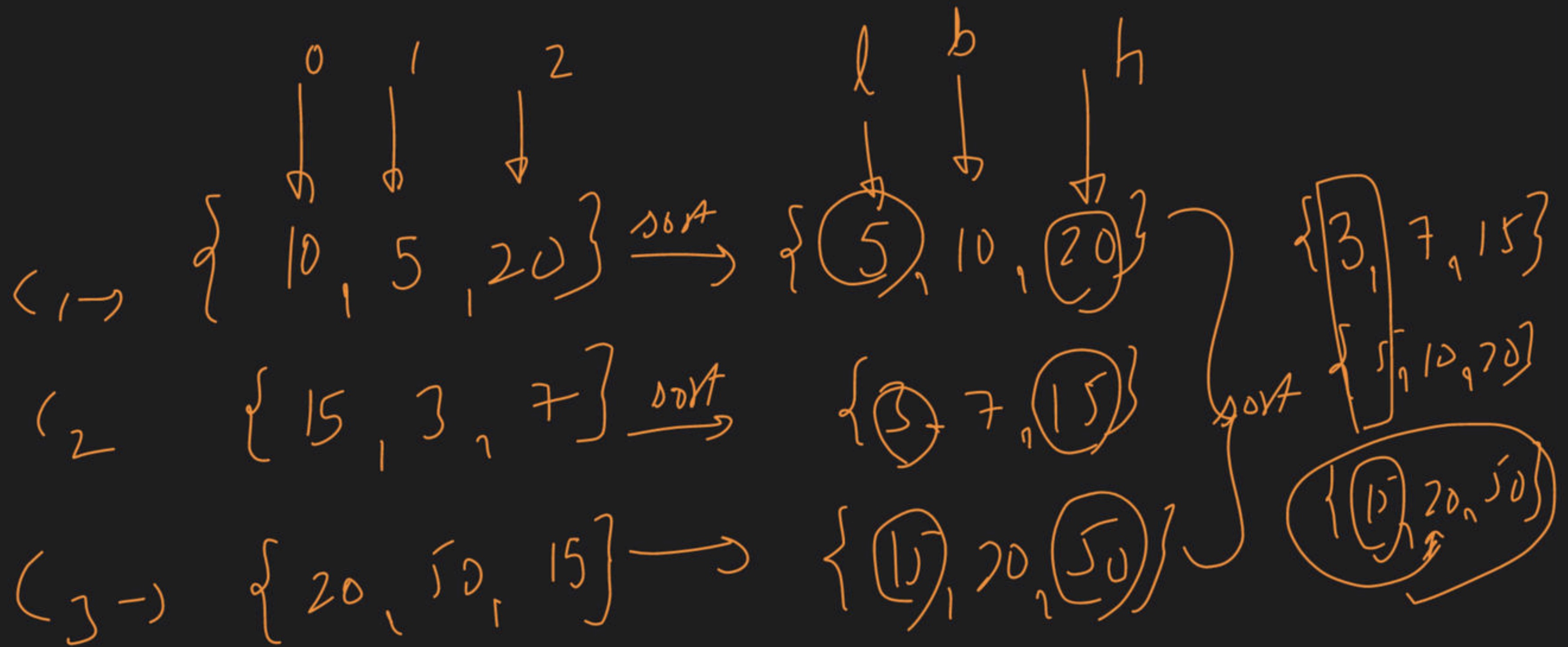
Longest Increasing Subsequence



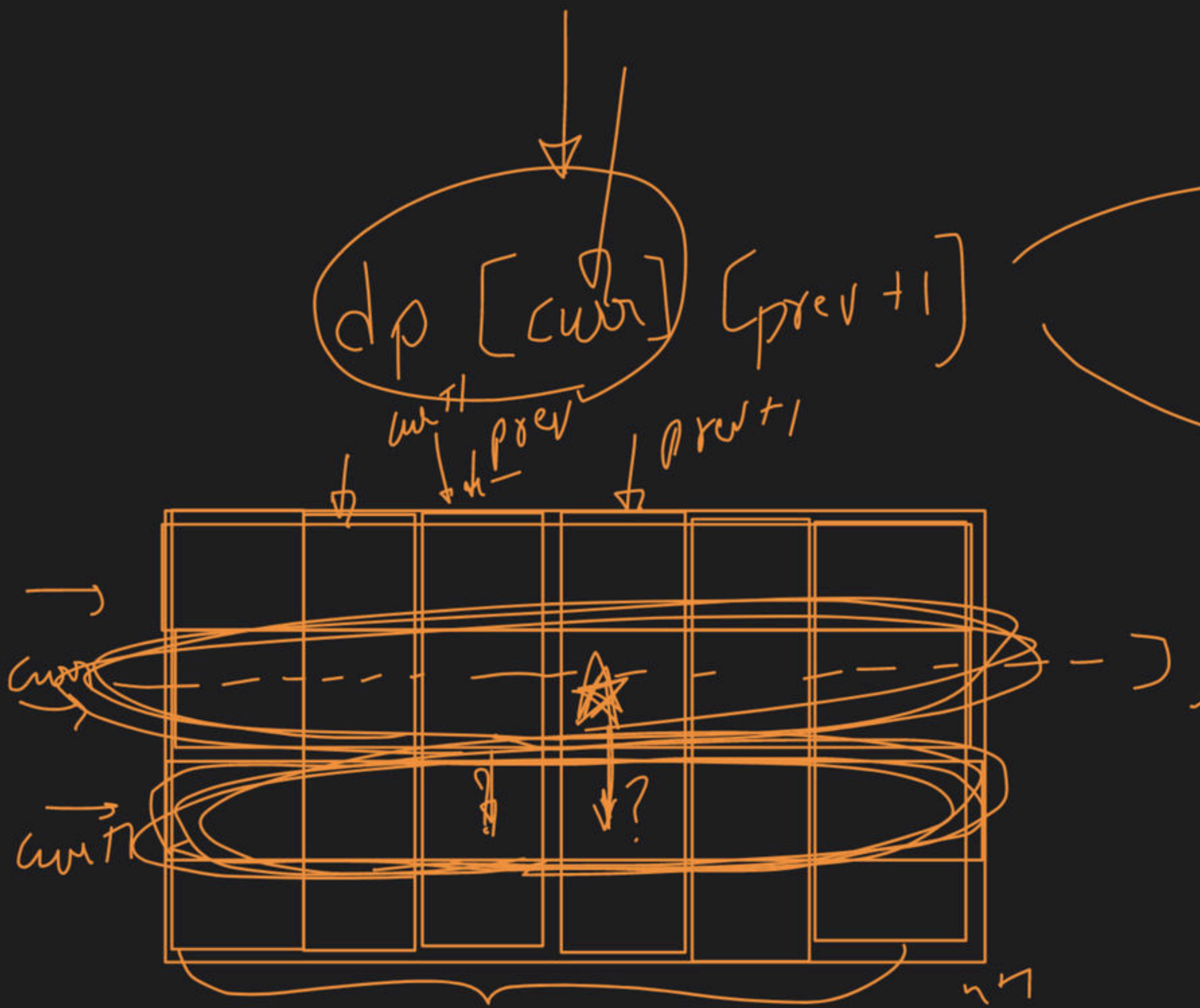


sort → $n \log n$

$\left\{ \begin{array}{ccc} l & b & h \\ - & - & - \\ \uparrow & \uparrow & \uparrow \end{array} \right\}$
, $O(1)$



max $\rightarrow$ height



$dp[cur][prev+1]$

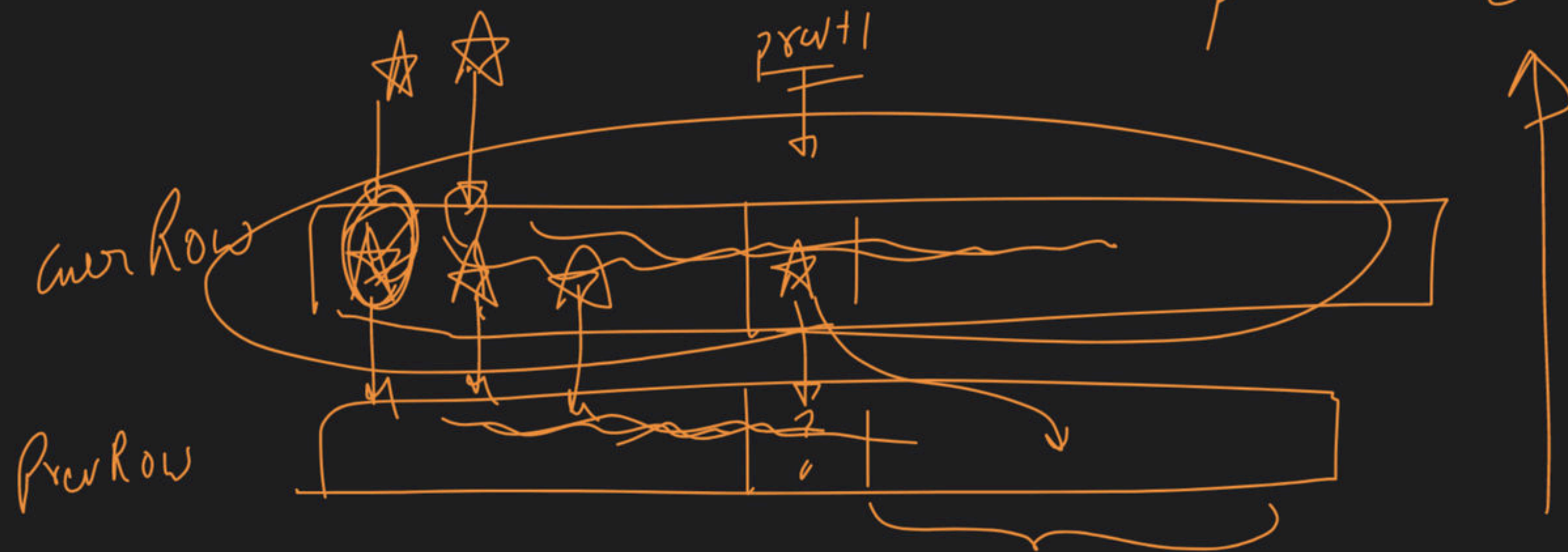


$dp[cur+1][prev+1]$



$dp[cur+1][cur+1]$

$curRow[prev+1]$ $\rightarrow prevRow[prev+1]$ ✓
 $\rightarrow prevRow[cur+1]$



?

←



x	y	z	z	z	+	+	-	-	-
---	---	---	---	---	---	---	---	---	---

cur row



